

Correlation and Clustering in Taiwan Stock Marke

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The 219×219 30-minute return correlation matrices of Taiwan stock market in the period of 2001-2003 are computed. The large eigenvalues of these correlation matrices indicate a high degree of collective motion of stock returns, while the distribution of small eigenvalues follows the random matrix theory prediction. The eigenvector components contain the information of industry clustering. Using eigenvector components, the minimal spanning trees can be constructed to reveal the global structure of Taiwan stock market.

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